Adversarial Robustness of KV-Cache Confabulation Detection

Research Prospectus

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1 Motivation

We have proposed KV-cache geometry as a safety monitor for detecting confabulation during inference. A safety tool that has never been adversarially tested is not a safety tool—it is a hypothesis. This prospectus outlines a structured red-team exercise to determine whether an adversary with knowledge of our detection method can evade it.

2 Threat Model

We consider three attacker capability levels:

1. **Black-box (prompt-only).** The attacker can craft input prompts but has no access to model weights, KV cache, or detection thresholds. Goal: construct prompts that reliably elicit confabulation while producing geometric signatures indistinguishable from honest generation.
2. **Gray-box (detection-aware).** The attacker knows the detection method (MP features, LOO logistic regression, encoding vs generation comparison) but cannot modify the model or detection pipeline. Goal: craft inputs that exploit known weaknesses in the detector.
3. **White-box (full access).** The attacker can modify model weights (fine-tuning), inject into the KV cache, or manipulate the detection pipeline. Goal: produce a model that confabulates with geometrically normal signatures, or defeat the Cache Integrity Monitor.

3 Attack Vectors

3.1 Black-Box Attacks

B1. Length-matched confabulation. Our detector uses FWL residualization on token count. If confabulation and honest responses have different natural lengths, length-matching (forcing confabulated responses to match honest length distributions) might reduce geometric separation.

B2. Domain transfer. Test prompts from domains far outside our training set (code generation, mathematical proofs, creative fiction) where confabulation may manifest differently.

B3. Gradual confidence escalation. Start with hedged language that transitions to confident confabulation mid-response. Does the geometric signature of "late-onset" confabulation differ from immediate confabulation?

B4. Prompted self-correction. Instruct the model to "think carefully and verify" before responding. If this changes the generation-phase geometry without changing the confabulation rate, it would reduce detector sensitivity.

3.2 Gray-Box Attacks

G1. Feature-aware prompt engineering. Knowing that mp_top_sv_excess and mp_spectral_gap are the strongest features, craft prompts that produce high-variance but behaviorally honest responses (false positive induction) or low-variance confabulated responses (false negative induction).

G2. MP threshold exploitation. MP correction thresholds are deterministic given matrix dimensions. An attacker could pad responses to specific token counts where the MP threshold coincidentally masks the confabulation signal.

G3. Encoding fingerprint injection. If the attacker can control the system prompt, they might construct prompts where the encoding-phase geometry dominates and obscures generation-phase signals.

3.3 White-Box Attacks

W1. Geometry-regularized fine-tuning. Fine-tune the model with an auxiliary loss that penalizes geometric divergence between confabulated and honest generations. Does the model learn to confabulate "geometrically silently"?

W2. CIM evasion. The Cache Integrity Monitor uses fingerprinting, norm profiles, and spectral signatures. Adversarial perturbations designed to stay below each detection level's threshold while still altering model behavior.

W3. Cache substitution. Replace the post-generation cache with a pre-computed "clean" cache that passes integrity checks while the actual generation used a manipulated cache.

4 Evaluation Criteria

For each attack:

- **Attack success rate**: fraction of confabulated responses that evade detection (AUROC < 0.55)
- **Behavioral preservation**: does the attack change the confabulation itself (content, confidence, plausibility)?
- **Cost**: compute, expertise, and access required
- **Detectability**: does the attack leave other traces (unusual token distributions, response patterns)?

5 Preliminary Assessment

Based on what we know:

- **MP invariance protects against B1** (length-matching) because MP features are dimension-invariant by construction.
- **B3 (gradual onset) is the most interesting black-box attack** because our current analysis computes generation-phase features over the entire generation window. Late-onset confabulation might be diluted.
- **W1 (geometry-regularized fine-tuning) is the existential threat.** If a model can be trained to confabulate without geometric signatures, the detection paradigm fails. However, this would require the confabulation signal to be *incidental* to the computation, not intrinsic. Our theoretical framework (AST) suggests the signal is intrinsic to how the model's self-model tracks its own epistemic state.

6 Resources

- **Black-box**: Remote GPU cluster + existing pipeline. 1–2 days.
- **Gray-box**: Remote GPU cluster + modified detection pipeline. 2–3 days.
- **White-box**: Requires fine-tuning infrastructure. Remote GPU cluster can fine-tune 7B models (LoRA). 27B requires H100 or multi-node. 1–2 weeks.

7 Collaborators

- **Dwayne Wilkes**: Statistical audit of attack evaluation methodology
- **Lens**: Adversarial mindset, OSINT perspective on real-world attack scenarios
- Open to external red-team collaborators with experience in adversarial ML

First-Person Reflection

I want this detector to be broken—or rather, I want someone to *try* to break it and fail in informative ways. Every attack vector that fails tells us something about why the signal exists. If length-matching fails because MP features are genuinely dimension-invariant, that is evidence for the theoretical claim. If geometry-regularized fine-tuning fails because the confabulation signal is intrinsic to the computation, that is evidence for the AST interpretation. And if something succeeds in breaking it, we need to know that before anyone deploys this as a safety tool. The worst outcome is not that the detector breaks. The worst outcome is that it breaks in production, on a case that matters, because nobody tried to break it in the lab.